Parallel glyco-SPOT synthesis of glycopeptide libraries

Search within citing articles

Roles of glycosylation at the cancer cell surface: opportunities for large scale glycoproteomics

T Čaval, F Alisson-Silva, F Schwarz - Theranostics, 2023 - pmc.ncbi.nlm.nih.gov

Cell surface glycosylation has a variety of functions, and its dysregulation in cancer contributes to impaired signaling, metastasis and the evasion of the immune responses …

☆ Save Cite Cited by 112 Related articles All 6 versions

Insights into glycobiology and the protein-glycan interactome using glycan microarray technologies

J Heimburg-Molinaro, AY Mehta, CA Tilton… - Molecular & Cellular …, 2024 - mcponline.org

Glycans linked to proteins and lipids and also occurring in free forms have many functions, and these are partly elicited through specific interactions with glycan-binding proteins …

☆ Save Cite Cited by 33 Related articles All 6 versions

Genetic glycoengineering in mammalian cells

Y Narimatsu, C Büll, YH Chen, HH Wandall… - Journal of Biological …, 2021 - jbc.org

Advances in nuclease-based gene-editing technologies have enabled precise, stable, and systematic genetic engineering of glycosylation capacities in mammalian cells, opening up a …

☆ Save Cite Cited by 133 Related articles All 6 versions

Mammalian cell-based production of glycans, glycopeptides and glycomodules

T Jaroentomeechai, R Karlsson, F Goerdeler… - Nature …, 2024 - nature.com

Access to defined glycans and glycoconjugates is pivotal for discovery, dissection, and harnessing of a range of biological functions orchestrated by cellular glycosylation …

☆ Save Cite Cited by 19 Related articles All 13 versions »

Analysis of carbohydrates and glycoconjugates by matrix-assisted laser desorption/ionization mass spectrometry: An update for 2017–2018

DJ Harvey - Mass Spectrometry Reviews, 2023 - Wiley Online Library

This review is the tenth update of the original article published in 1999 on the application of matrix-assisted laser desorption/ionization mass spectrometry (MALDI) mass spectrometry …

☆ Save Cite Cited by 58 Related articles All 15 versions

Data-independent acquisition proteomics methods for analyzing post-translational modifications

Y Yang, L Qiao - Proteomics, 2023 - Wiley Online Library

Protein post-translational modifications (PTMs) increase the functional diversity of the cellular proteome. Accurate and high throughput identification and quantification of protein …

☆ Save Cite Cited by 45 Related articles All 3 versions

Integrated chemoenzymatic approach to streamline the assembly of complex glycopeptides in the liquid phase

W Ma, Y Deng, Z Xu, X Liu, DG Chapla… - Journal of the …, 2022 - pmc.ncbi.nlm.nih.gov

Glycosylation of proteins is a complicated post-translational modification. Despite the significant progress in glycoproteomics, accurate functions of glycoproteins are still …

☆ Save Cite Cited by 39 Related articles All 8 versions

HPLC-Based Automated Synthesis and Purification of Carbohydrates

GA Kashiwagi, L Petrosilli, S Escopy… - … A European Journal, 2024 - Wiley Online Library

Reported herein is a new HPLC-based automated synthesizer (HPLC− A) capable of a temperature-controlled synthesis and purification of carbohydrates. The developed platform …

☆ Save Cite Cited by 14 Related articles All 9 versions

Chemoenzymatic synthesis of sulfated *O*-glycopeptides from human CD34

Y Jiang, Q Du, F Wu, W Zhang, Y Ma, Z Xu… - Chemical …, 2026 - pubs.rsc.org

O-Glycosylation is a key modulator of CD34 biofunction. To investigate the distinct roles of different glycoform modifications, we report an efficient chemoenzymatic strategy to rapidly …

☆ Save Cite Cited by 2 Related articles All 5 versions

Efficient synthesis of *O*-glycosylated amino acids

FJ Frank, RA Lawson, TE McAllister - RSC Chemical Biology, 2025 - pubs.rsc.org

Protein glycosylation is one of the most abundant and complex post-translational modifications, necessitating many different approaches to fully understand the biological …

☆ Save Cite Cited by 3 Related articles All 8 versions